SWATI AGGRAWAL

swatiagg357@gmail.com | +91-7011930926 | github.com/swatiaggrawal | linkedin.com/in/swatiaggrawal

PROFESSIONAL SUMMARY

M.Sc. Computer Science graduate from Sapienza University of Rome with a strong foundation in applied AI/ML and computer vision, including Vision Transformers, CNNs, transfer learning, and continual learning. Experienced in developing end-to-end ML pipelines using Python, PyTorch, TensorFlow, FastAPI, Docker, and AWS. Research background in computer vision and environmental machine learning, including an IEEE publication on ML approaches for air-quality evaluation and prediction covering regional, remote-sensing, and sensor-based applications. Currently applying this foundation to geospatial AI and remote-sensing problems.

RELEVANT SKILLS

- **AI & Machine Learning:** Deep Learning, Computer Vision, Vision Transformers, Transfer Learning, Continual Learning, PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face Transformers
- **Environmental & Geospatial Exposure:** Air-quality ML, environmental data, regional air-quality modelling, remote sensing applications, satellite-derived data
- **Software Engineering:** Python, C++, Java, SQL, DSA, OOP, FastAPI, REST APIs, React, Git
- **Cloud & DevOps:** Docker, AWS (ECS, Fargate, CloudWatch), Azure, Linux, Render
- **Data Science:** NumPy, Pandas, Matplotlib

EDUCATION

M.Sc. Computer Science Sapienza Università di Roma *2026*

- Coursework: Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- Thesis: Evaluating Pretrained Vision Transformers Under Continual Learning using adapter-based methods with Elastic Weight Consolidation (EWC) for image classification

B.Tech, CSE with AI and ML Sharda University *2023*

- Coursework: Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- Awarded merit-based scholarship (Top 5–10% of cohort)

EXPERIENCE

Researcher at Vision Lab Sapienza Università di Roma (Master's Thesis) *Nov 2025 – May 2026*
https://github.com/swatiaggrawal/Thesis-Vision-Transformers

- Conducted research under Prof. Marco Raoul Marini on continual learning for computer vision, developing image-classification pipelines using pretrained Vision Transformers and transfer-learning techniques.
- Designed adapter-based learning and EWC techniques, achieving 90.59% average accuracy with only 2.01% average forgetting on Split CIFAR-100 across 5 sequential tasks, demonstrating substantial reduction in catastrophic forgetting.
- Benchmarked against ResNet-18 and naive ViT fine-tuning baselines (24.22% and 41.91% average accuracy respectively), validating results via cross-seed evaluation.
- Technologies: Python, PyTorch, Vision Transformers, Computer Vision

PUBLICATION

Machine Learning for Air Pollution IEEE Conference Publication, 2023
https://doi.org/10.1109/ViTECoN58111.2023.10157028

- Reviewed machine learning approaches for air-quality evaluation and prediction, including Random Forest, SVM, Decision Trees, ANN, and clustering methods across environmental and regional applications.
- Examined applications of remote sensing, satellite-derived AOD data, IoT sensing networks, and localized air-quality monitoring for environmental prediction.

SELECTED PROJECTS

Metal Defect Detection using Computer Vision
Code: https://github.com/swatiaggrawal/Metal-Defect-Detection

- Built a custom CNN using TensorFlow/Keras for automated image-based defect classification across 6 classes, achieving 97.22% test accuracy.

Knowledge Copilot: Grounded RAG Chatbot
Code: https://github.com/swatiaggrawal/RAG-Chatbot
Live demo: https://rag-chatbot-new.onrender.com

- Built an end-to-end RAG chatbot with a modular 4-stage pipeline (ingestion, embedding, retrieval, generation); integrated Gemini API embeddings with Pinecone vector store; deployed within a 512MB free-tier RAM limit.

DocDelta: AI-Powered Document Intelligence & Comparison Platform
Code: https://github.com/swatiaggrawal/DocDelta

- Developed a semantic document comparison API using FastAPI and Sentence Transformers; containerized with Docker and deployed on AWS ECS/Fargate with CloudWatch monitoring.